Curriculum Vitae

D.Sc. (Tech.) Xin Liu

- Name: Liu, Xin
- ORCiD: 0000-0002-2242-6139
- ResearcherID: AAD-5166-2019
- Scopus ID: 57192258303
- Google Scholar: https://scholar.google.com/citations?user=xHkC5U0AAAAJ
- Webpage: https://linuxsino.github.io/
- Date of CV: 04.11.2026

Degrees

- 03.2019, Doctor of Science (Technology), major subject: Computer Science and Engineering, University of Oulu, Oulu, Finland. Supervisors: Prof. Guoying Zhao (Member of Academia Europaea, Member of Finnish Academy of Sciences and Letters, Fellow of IEEE, IAPR, ELLIS and AAIA) and Prof. Matti Pietikäinen (IEEE Life Fellow)
- 09.2016, Doctor of Engineering, major subject: Information and Communication Engineering, Xi'an Jiaotong University, Xi'an, China. Supervisor: Prof. Chun Qi
- 04.2007, Master of Engineering, major subject: Computer Science, Kunming University of Science and Technology, Kunming, China
- 07.2003, Bachelor of Engineering, major subject: Computer Science, Changchun University of Science and Technology, Changchun, China

Current Employment

- 01.2026–present, Associate Professor (Tenured), Shanghai Jiao Tong University, Finland

Previous Work Experience

- 08.2021–07.2025, Associate Professor (tenure track), Lappeenranta-Lahti University of Technology, Lappeenranta, Finland
- 08.2020–07.2021, Associate Professor, Tianjin University, Tianjin, China
- 04.2019–07.2020, Senior Researcher, University of Oulu, Oulu, Finland
- 08.2019–09.2019, Visiting Scholar, Aalto University (Prof. Juho Kannala), Helsinki, Finland
- 03.2019–08.2019, Visiting Scholar, Imperial College London (Prof. Stefanos Zafeiriou), London, United Kingdom
- 08.2017–02.2018, Endeavour Research Fellow, University of Sydney (Prof. Dacheng Tao), Sydney, Australia
- 05.2014–03.2019, Doctoral Researcher, University of Oulu, Oulu, Finland
- 04.2007–11.2010, Co-founder, CTO, and Vice President, HR-SkyEyes Ltd., Shenzhen, China

Research Funding & Grants

- 11.2023–12.2024, "Large Multimodal Model for Human Emotion Anticipation", **LUMI super computing project**, **CSC – IT Center for Science**, Finland, **PI**
- 09.2020–08.2023, "Win or lose? Emotion AI via identity-free body language recognition and understanding", **Postdoctoral Researcher**, **Academy of Finland**, 252 980 €, **PI**
- 03.2019–08.2019, "Human micro-gesture analysis for hidden emotion understanding", **KAUTE Grant, KAUTE Foundation**, 5 000 €, **PI**
- 06.2017–01.2018, "Communicative Gesture Analysis using Computer Vision Methods", **Tekniikan edistämissäätiö (TES) Grant**, **Finnish Foundation for Technology Promotion**, 5 000 €, **PI**
- 08.2017–02.2018, "Communicative Gesture Analysis using Temporal Dynamics and Context Understanding", **Endeavour Research Fellowship, Australian Government Department of Education and Training**, 25 000 AU$, **PI**

- 07.2016–12.2016, "Communicative Gesture Analysis using Computer Vision Methods", **Otto A. Malm Grant**, **Otto A. Malm Foundation**, 3 500 €, **PI**
- 03.2016–07.2016, "Analyzing Communicative Gestures using Machine Vision Methods", **Tauno Tönning Research Grant**, **Tauno Tönning Research Foundation**, 3 500 €, **PI**
- 01.2015–07.2015, "Communicative Gesture Analysis using Computer Vision Methods", **Nokia Scholarship**, **Nokia Foundation**, 5 000 €, **PI**
- 09.2018–10.2020, "Microgesture analysis with machine learning for hidden emotion understanding (MiGA)", **Academy project**, **Academy of Finland**, 857 145 €, Key member, PI: Guoying Zhao
- 09.2014–12.2017, "Computer vision methods for analyzing human communicative behavior", **Academy fellow position funding**, **Academy of Finland**, 391 478 €, Key member, PI: Guoying Zhao

Research Output

Selected publications since 2020:

1. Rong Gao, Xinrui Li, Deng Li, **Xin Liu***, Heikki Kälviäinen. "Speak on Cue, Remember on Point: Event-Aware Streaming Video Commentary." ***ACM International Conference on Multimedia (ACMMM)***, 2026.
2. Bohao Xing, Zitong Yu, **Xin Liu***, Qilang Ye, Weicheng Xie, Huanjing Yue, Jingyu Yang, and Heikki Kälviäinen. "EMO-LLaMA: Enhancing Facial Emotion Understanding with Instruction Tuning." ***International Journal of Computer Vision (IJCV)***, 2026.
3. Bohao Xing, **Xin Liu***, Guoying Zhao, Chengyu Liu, Xiaolan Fu, and Heikki Kälviäinen, "EmotionHallucer: Evaluating Emotion Hallucinations in Multimodal Large Language Models." ***International Conference on Learning Representations (ICLR)***. 2026.
4. Kaishen Yuan, Zitong Yu, **Xin Liu***, Bohao Xing, Yuting Zhang, Weicheng Xie, Linlin Shen, Björn Schuller. "Multi-granularity Facial Emotional Representation with Unlabeled Data and Textual Supervision." ***IEEE Transactions on Image Processing (TIP)***, 2026.
5. Huanjing Yue, Xianghui Li, Mengyuan Sun, Gaosheng Liu, **Xin Liu**, Jingyu Yang. "DIC-DDA: Learned Distributed Image Compression via Dual Domain Alignment." ***IEEE Transactions on Image Processing (TIP)***, 2026.
6. Deng Li, Bohao Xing, **Xin Liu***, Baiqiang Xia, Bihan Wen, and Heikki Kälviäinen. "DEEMO: De-identity Multimodal Emotion Recognition and Reasoning." ***ACM International Conference on Multimedia (ACMMM)***, 2025.
7. Zhuozhao Hu, Kaishen Yuan, **Xin Liu***, Zitong Yu, Yuan Zong, Jingang Shi, Huanjing Yue, and Jingyu Yang. "FEALLM: Advancing Facial Emotion Analysis in Multimodal Large Language Models with Emotional Synergy and Reasoning."***ACM International Conference on Multimedia (ACMMM)***, 2025.
8. Deng Li, Jun Shao, Bohao Xing, Rong Gao, Bihan Wen, Heikki Kälviäinen, and **Xin Liu***. "MSF-Mamba: Motion-aware State Fusion Mamba for Efficient Micro-Gesture Recognition." ***IEEE Transactions on Multimedia (TMM)***. 2025.
9. Chao Hao, Zezheng Wang, Yanhua Huang, Ruiwen Xu, Wenzhe Niu, **Xin Liu**, and Zitong Yu. "Dynamic Collaboration of Multi-Language Models based on Minimal Complete Semantic Units." ***Empirical Methods in Natural Language Processing (EMNLP)***. 2025.
10. Jingyi Yang, Xun Lin, Zitong Yu, Liepiao Zhang, **Xin Liu**, Hui Li, Xiaochen Yuan, Xiaochun Cao. "DADM: Dual Alignment of Domain and Modality for Face Anti-spoofing." ***International Conference on Computer Vision (ICCV)***. 2025
11. Qilang Ye, Zitong Yu, Rui Shao, Yawen Cui, Xiangui Kang, **Xin Liu**, Philip Torr, and Xiaochun Cao. "CAT+: Investigating and Enhancing Audio-visual Understanding in Large Language Models." ***IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)***. 2025
12. Rong Gao, **Xin Liu***, Zhuozhao Hu, Bohao Xing, Baiqiang Xia, Zitong Yu, and Heikki Kälviäinen. "FSBench: A Figure Skating Benchmark for Advancing Artistic Sports Understanding." ***IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)***. 2025.

13. Yuting Zhang, Hao Lu, Qingyong Hu, Yin Wang, Kaishen Yuan, **Xin Liu**, and Kaishun Wu. “Period-LLM: Extending the Periodic Capability of Multimodal Large Language Model.” ***IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)***. 2025.
14. Yuting Zhang, Hao Lu, **Xin Liu***, Yingcong Chen, and Kaishun Wu, "Advancing Generalizable Remote Physiological Measurement through the Integration of Explicit and Implicit Prior Knowledge." ***IEEE Transactions on Image Processing (TIP)***. 2025.
15. Chao Hao, Zitong Yu, **Xin Liu***, Jun Xu, Huanjing Yue, and Jingyu Yang, “A Simple yet Effective Network based on Vision Transformer for Camouflaged Object and Salient Object Detection.” ***IEEE Transactions on Image Processing (TIP)***. 2025. (**ESI highly cited paper**)
16. Cong Cao, Huanjing Yue, **Xin Liu**, and Jingyu Yang, “Zero-shot Video Restoration and Enhancement Using Pre-Trained Image Diffusion Model. ” ***AAAI Conference on Artificial Intelligence (AAAI)***. 2025.
17. **Xin Liu**, Kaishen Yuan, Xuesong Niu, Jingang Shi, Zitong Yu, Huanjing Yue, and Jingyu Yang. “Multi-scale Promoted Self-adjusting Correlation Learning for Facial Action Unit Detection.” ***IEEE Transactions on Affective Computing (TAFFECT)***. 2024.
18. Xinxu Ge, **Xin Liu***, Zitong Yu, Jingang Shi, Chun Qi, Jie Li, and Heikki Kälviäinen, “DiffFAS: Face Anti-Spoofing via Generative Diffusion Models.” ***European Conference on Computer Vision (ECCV)***. 2024.
19. Kaishen Yuan, Zitong Yu, **Xin Liu***, Weicheng Xie, Huanjing Yue, and Jingyu Yang, “AUFormer: Vision Transformers are Parameter-Efficient Facial Action Unit Detectors.” ***European Conference on Computer Vision (ECCV)***. 2024.
20. Zitong Yu, Rizhao Cai, Yawen Cui, **Xin Liu**, Yongjian Hu and Alex C. Kot. “Rethinking Vision Transformer and Masked Autoencoder in Multimodal Face Anti-Spoofing.” ***International Journal of Computer Vision (IJCV)***. 2024.
21. **Xin Liu**, Yuting Zhang, Zitong Yu, Hao Lu, Huanjing Yue, and Jingyu Yang. “rPPG-MAE: Self-supervised Pre-training with Masked Autoencoders for Remote Physiological Measurement.” ***IEEE Transactions on Multimedia (TMM)***. 2024.
22. Huanjing Yue, Yijia Cheng, **Xin Liu**, and Jingyu Yang. “Recaptured Raw Screen Image and Video Demoiréing via Channel and Spatial Modulations.” ***The Thirty-Seventh Annual Conference on Neural Information Processing Systems (NeurIPS)***. 2023.
23. Haoyu Chen, Henglin Shi, **Xin Liu**, Xiaobai Li, and Guoying Zhao. “SMG: A Micro-Gesture Dataset Towards Spontaneous Body Gestures for Emotional Stress State Analysis.” ***International Journal of Computer Vision (IJCV)***. 2023.
24. Zitong Yu, Benjia Zhou, Jun Wan, Pichao Wang, Haoyu Chen, **Xin Liu**, Stan Z. Li, and Guoying Zhao. “Searching multi-rate and multi-modal temporal enhanced networks for gesture recognition.” ***IEEE Transactions on Image Processing (TIP)***, 2021.
25. **Xin Liu**, Henglin Shi, Haoyu Chen, Zitong Yu, Xiaobai Li, Guoying Zhao, “iMiGUE: An Identity-free Video Dataset for Micro-Gesture Understanding and Emotion Analysis”, ***IEEE Conference on Computer Vision and Pattern Recognition (CVPR)***, 2021.
26. **Xin Liu**, Guoying Zhao. “3D Skeletal Gesture Recognition via Discriminative Coding on Time-Warping Invariant Riemannian Trajectories”, ***IEEE Transactions on Multimedia (TMM)***, 2021.
27. Haoyu Chen, **Xin Liu**, Jingang Shi, Guoying Zhao. "Temporal hierarchical dictionary guided decoding for online gesture segmentation and recognition." ***IEEE Transactions on Image Processing (TIP)***, 2020.
28. **Xin Liu**, Henglin Shi, Xiaopeng Hong, Haoyu Chen, Dacheng Tao, Guoying Zhao, “3D Skeletal Gesture Recognition via Hidden States Exploration”, ***IEEE Transactions on Image Processing (TIP)***, 2020.

Other research outputs:

- **LSD** (https://github.com/linuxsino/LSD), a toolbox for video background subtraction: **Xin Liu**, Guoying Zhao, Jiawen Yao, Chun Qi. “Background Subtraction Based on Low-rank and Structured Sparse Decomposition”, ***IEEE Transactions on Image Processing***, 2015. (**ESI highly cited paper**)
- **20 Patents** (China) related to machine learning and computer vision

Research Supervision & Leadership Experience

- Deng Li, "Emotional Intelligence...", Doctoral thesis, Lappeenranta-Lahti University of Technology LUT, 2022–2026, Principal supervisor
- Rong Gao, "Human Emotion Analysis...", Doctoral thesis, Lappeenranta-Lahti University of Technology LUT, 2023–, Supervisor
- Bohao Xing, " Advancing Secure and Privacy-Preserving Emotion AI...", Doctoral thesis, Lappeenranta-Lahti University of Technology LUT, 2024–, Supervisor
- Henglin Shi, "Learning-based human...", Doctoral thesis, University of Oulu, 2016-2022, Co-supervisor
- Haoyu Chen, "Human gesture...". Doctoral thesis, University of Oulu, 2017-2022, Co-supervisor
- Supervisor of undergraduate students' theses: 20 (M.Sc.), 12 (B.Sc.)
- CTO & Co-founder, HR-SkyEyes Ltd. (Apr 2007 – Nov 2010): Led an R&D team of 10+ researchers

Teaching Merits

- Taught at Lappeenranta-Lahti University of Technology LUT at the master and postgraduate level from 2021. The number of courses: 2; the number of course implementations: 7
- Taught at Tianjin University at the bachelor level from 2020. The number of courses: 1; the number of course implementations: 2
- Teaching assistance of 3 courses at Xi'an Jiaotong University, and 1 course at the University of Oulu

Awards & Honours

- **Second prize of the IEEE Finland Jt. Chapter SP/CAS Best Paper Award 2022.** Zitong Yu, Benjia Zhou, Jun Wan, Pichao Wang, Haoyu Chen, **Xin Liu**, Stan Z. Li, and Guoying Zhao. "Searching multi-rate and multi-modal temporal enhanced networks for gesture recognition." ***IEEE Transactions on Image Processing (TIP)***, 2021.
- **Runner Up**. Haoyu Chen, Zitong Yu, **Xin Liu**, et al., "2nd Place Scheme on Action Recognition Track of ECCV 2020 VIPriors Challenges: An Efficient Optical Flow Stream Guided Framework", **ECCV 2020 VIPriors Challenges**, ***European Conference on Computer Vision (ECCV)***, 2020.
- **Best Paper Award**. Zhiyuan Zha, **Xin Liu**, Xiaohua Huang, et al., "Analyzing the group sparsity based on the rank minimization methods," ***IEEE International Conference on Multimedia and Expo (ICME)***, 2017.
- **Endeavour Research Fellowship, Endeavour Leadership Program of Australian Government Department of Education and Training,** 2016.

Editorial and Reviewing Activities

- Editorial board member/Guest editor:
  i. ***IEEE Transactions on Multimedia****, 2025-now*
  ii. ***Computing and Artificial Intelligence***, 2024-now
  iii. ***Artificial Intelligence and Applications***, 2023-2024
  iv. Guest editor of "Application of Affective Computing" on ***Applied Sciences***, 2023-2025

- Workshop/Special session (co-) organizer:
  i. Subtle Visual Computing Workshop & 1st Multimodal Deception Detection Competition with ***ACMMM 2025***
  ii. MER 2025: When Affective Computing Meets Large Language Models with ***ACMMM 2025***
  iii. Special Session of Multimodal Human Behavior Understanding and Generation with ***IEEE IJCB 2024***
  iv. 1st Workshop & Challenge on Micro-gesture Analysis for Hidden Emotion Understanding with ***IEEE IJCAI 2023***

- Peer reviewer of funding applications: 26 times of National Natural Science Foundation of China
- Pre-examiner of the doctoral dissertations: 8 times in Finnish and Chinese universities
- Area Chair or Senior Program Committee:

i. AAAI 2027
ii. *AAAI 2026* (AI Alignment Track)
iii. ICPR 2026
iv. International Conference on Advances in Signal, Image and Video Processing (*SIGNAL*) 2018, 2019, 2020, 2021

- Referee for journals: e.g., *IEEE Transactions on Pattern Analysis and Machine Intelligence, International Journal of Computer Vision, Proceedings of the IEEE, IEEE Transactions on Image Processing, IEEE Transactions on Signal Processing, IEEE Transactions on Information Forensics & Security, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Cybernetics, IEEE Transactions on Circuits and Systems for Video Technology, IEEE Transactions on Computational Imaging, IEEE Transactions on Multimedia, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Computational Social Systems, IEEE Transactions on Medical Imaging, IEEE Transactions on Network Science and Engineering, IEEE Transactions on Consumer Electronics, IEEE Transactions on Intelligent Vehicles, IEEE Transactions on Big Data, IEEE Transactions on Circuits and Systems II: Express Briefs, IEEE Transactions on Human-Machine Systems, IEEE Transactions on Signal and Information Processing over Networks, IEEE Transactions on Industrial Informatics, IEEE Transactions on Mobile Computing, IEEE Transactions on Emerging Topics in Computational Intelligence, IEEE Transactions on Biomedical Engineering, IEEE Transactions on Neural Systems & Rehabilitation Engineering, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Cognitive and Developmental Systems, IEEE Transactions on Geoscience and Remote Sensing, IEEE Transactions on Biometrics, Behavior and Identity Science, IEEE Internet of Things Journal, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, IEEE Geoscience and Remote Sensing Magazine, IEEE Journal of Biomedical and Health Informatics, IEEE Multimedia, IEEE Pervasive Computing, IEEE Open Journal of Signal Processing, AAAS RESEARCH, Pattern Recognition, Neural Networks, ACM Transactions on Multimedia Computing, Communications, and Applications, IEEE Signal Processing Letters, Digital Signal Processing, Science Progress, Neurocomputing, Knowledge-Based Systems, IEEE Access, Artificial Intelligence Review, SPIE Journal of Electronic Imaging, Soft Computing, Visual Computer, IET Electronics Letters, Multimedia Tools and Application, PlosONE, Journal of Computational Design and Engineering, Computers in Biology and Medicine*
- Referee or PC for conferences: e.g., *CVPR 2024-2026, ICCV 2023, 2025, ECCV 2024, 2026, NeurIPS 2024-2026, ICML 2025, 2026, ICLR 2025-2027, AISTATS 2025, 2026, AAAI 2026, 2027, ACMMM 2024-2026, FG 2018, WACV 2024, 2025, BMVC 2017, IJCB 2024, ACCV 2016, SCIA 2023, 2025*

Other Key Academic Merits

- IEEE Senior Member, ELLIS Member, ACM Member, SPIE Member, APA Member
- Citations: 3610 (Google Scholar); 2077 (Scopus); 2581 (Web of Science)
- H-index: 34 (Google Scholar); 25 (Scopus); 28 (Web of Science)